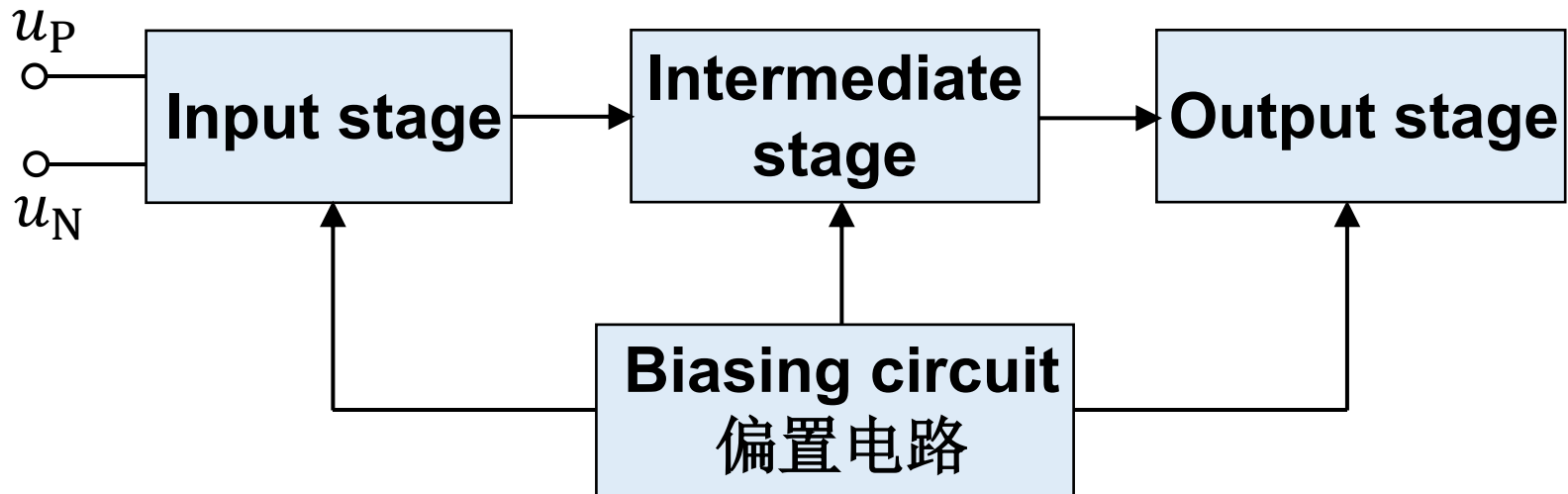


4.4 Constant current source circuit 电流源电路

The structure of the integrated operational circuit (IOC)

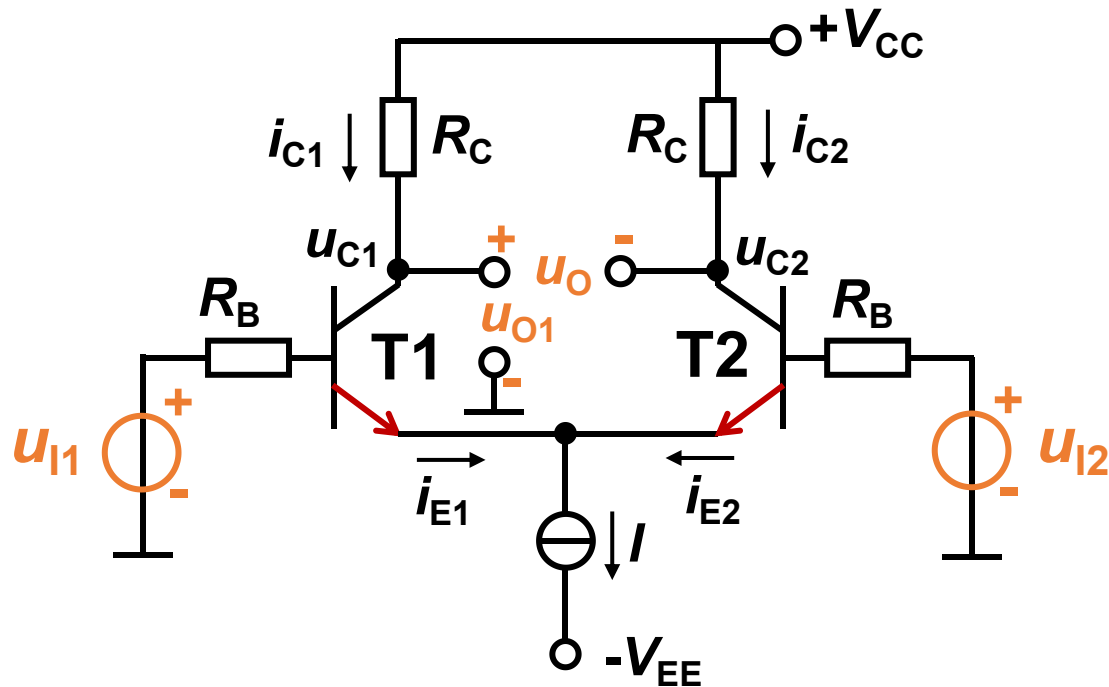


Biasing circuit: provide the proper quiescent working points of all amplifier circuits

In IOC, constant current sources are chosen as biasing circuits

4.4.1 Differential amplifier with constant-current source

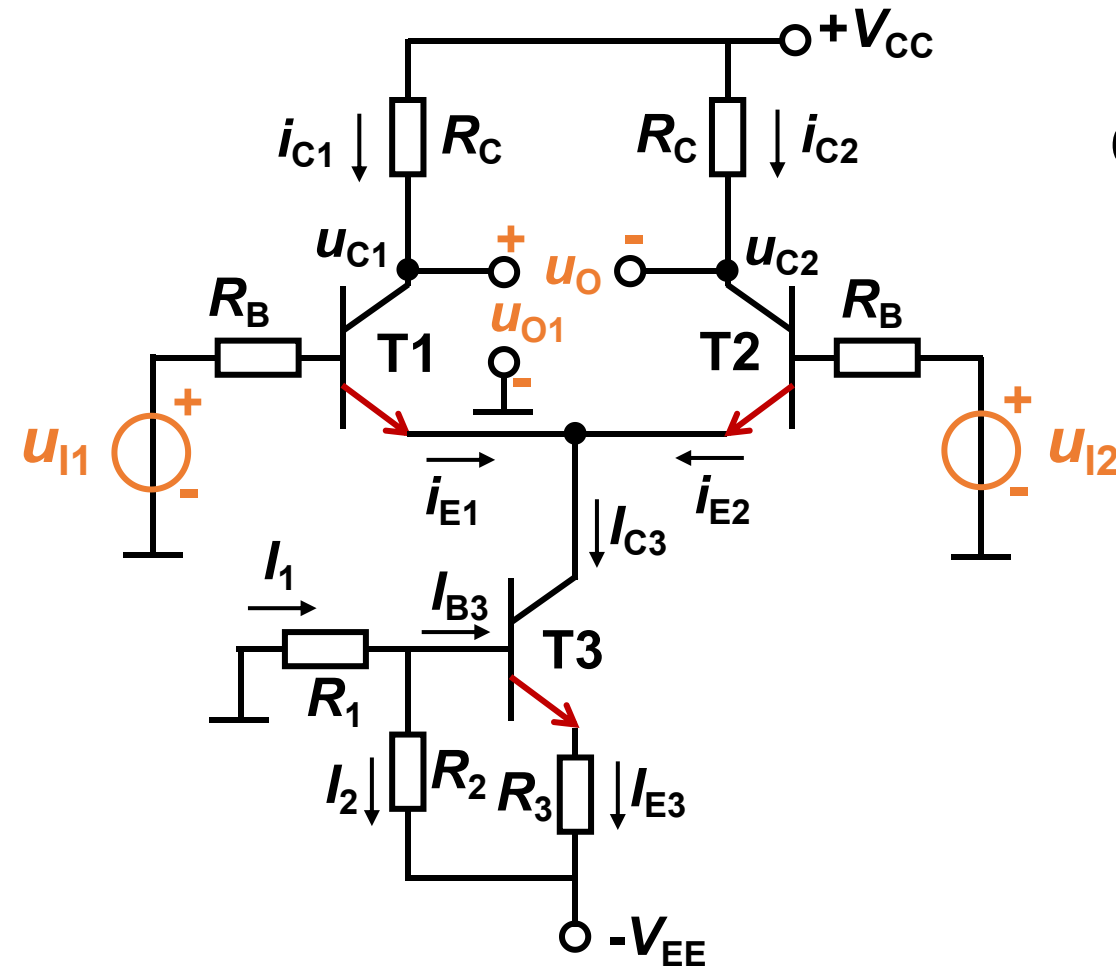
帶恒流源的差分放大电路



Constant-current source:

- 1) Very high AC resistance $\rightarrow$ High K_{CMR}
- 2) Low DC resistance $\rightarrow$ Low V_{EE}

The DC current is important for the constant-current source



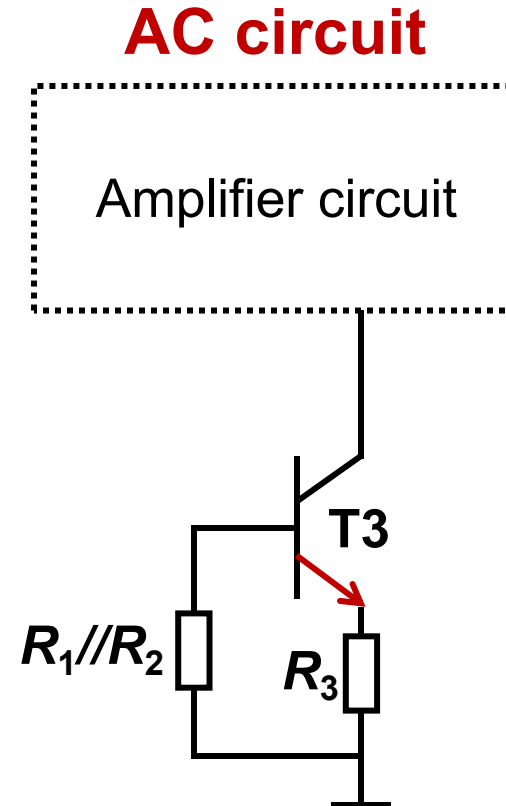
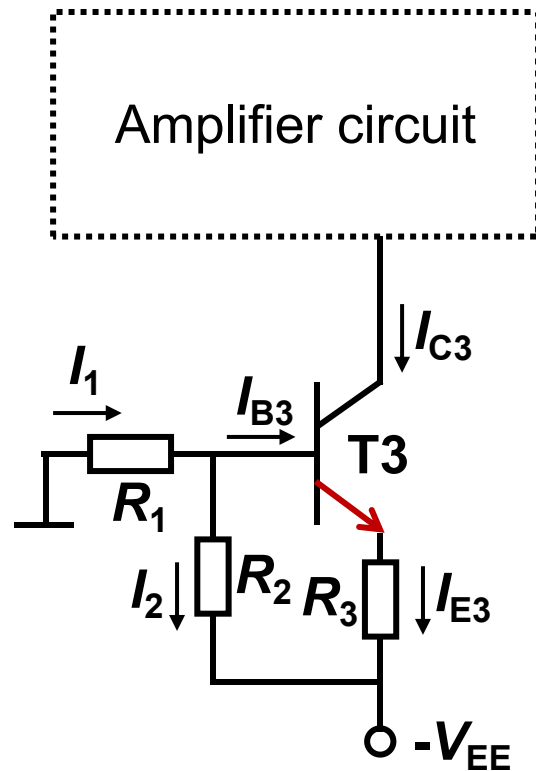
Choose R_1 , R_2 and R_3 ,
let $I_2 \gg I_{B3}$, $I_2 \approx I_1$

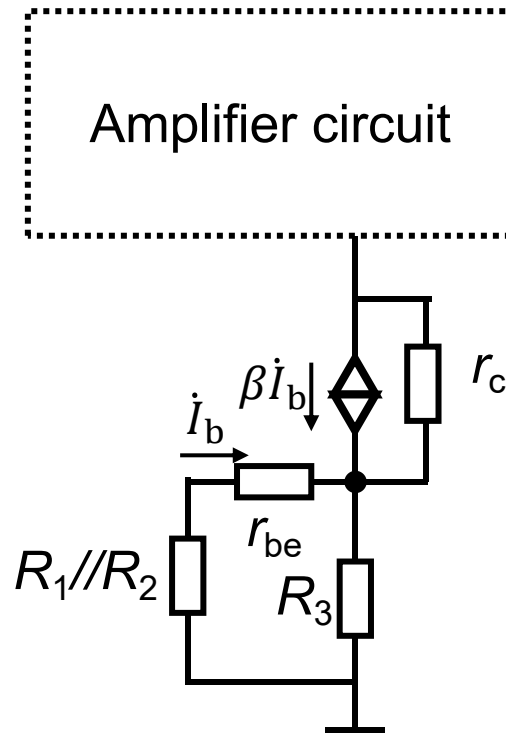
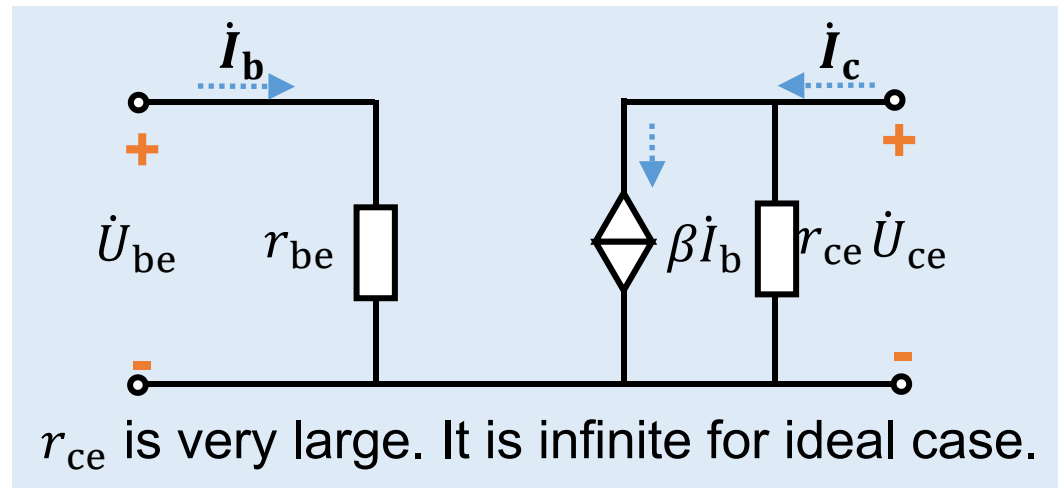
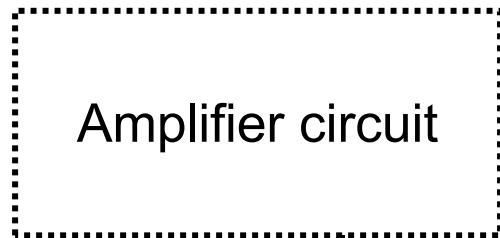
The voltage drop on R_2 :

$$U_{R2} \approx V_{EE} \frac{R_2}{R_1 + R_2}$$

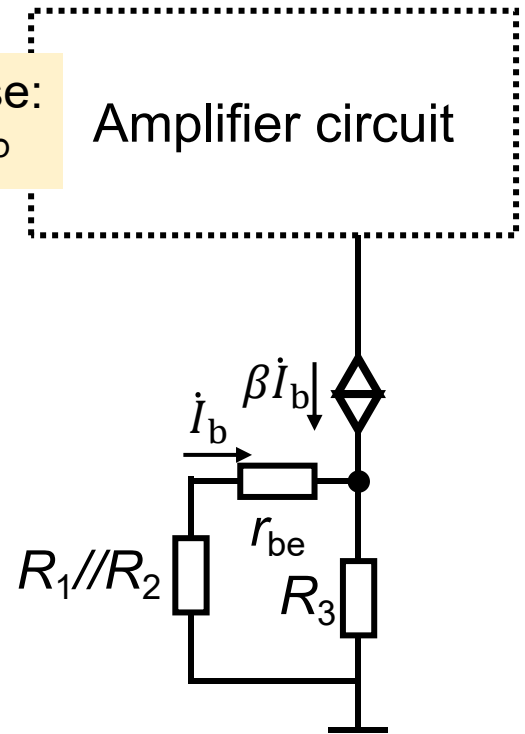
$$I_{C3} \approx I_{E3} = \frac{U_{R2} - U_{BE3}}{R_3}$$

For constant current source, we mainly focus on its DC circuit and quiescent working parameters. This is because its AC resistance is very large.

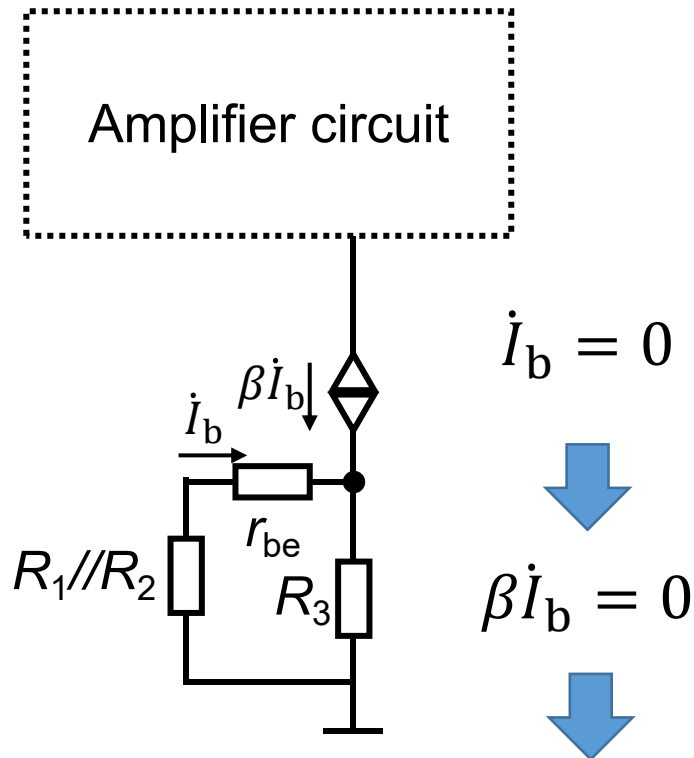




Ideal case:
 $r_{ce} \rightarrow \infty$

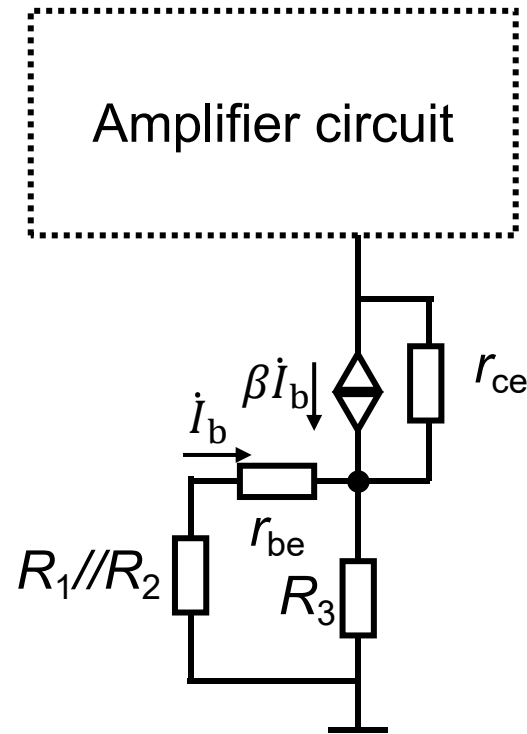


Ideal case: $r_{ce} \rightarrow \infty$



The current source has infinite AC resistance.

Real case: r_{ce}



The current source has a finite but very large AC resistance.

4.4.2 Current mirror 镜像电流源

T0 and T1 are identical

$$\beta_0 = \beta_1 = \beta$$

$$\therefore U_{CE0} = U_{BE0}$$

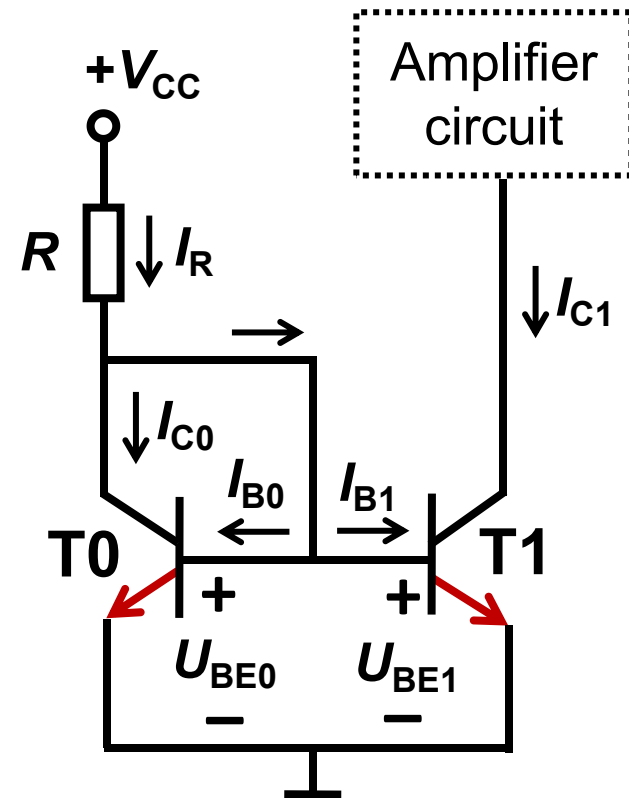
$\therefore$ T0 is always working in amplification region

$$I_{C0} = \beta I_{B0}$$

$$\therefore U_{BE0} = U_{BE1}$$

$$\therefore I_{B0} = I_{B1} = I_B$$

$$\therefore I_{C1} = \beta I_B = I_{C0}$$



I_{C0} and I_{C1} are mirror symmetric

The reference current 参考电流:

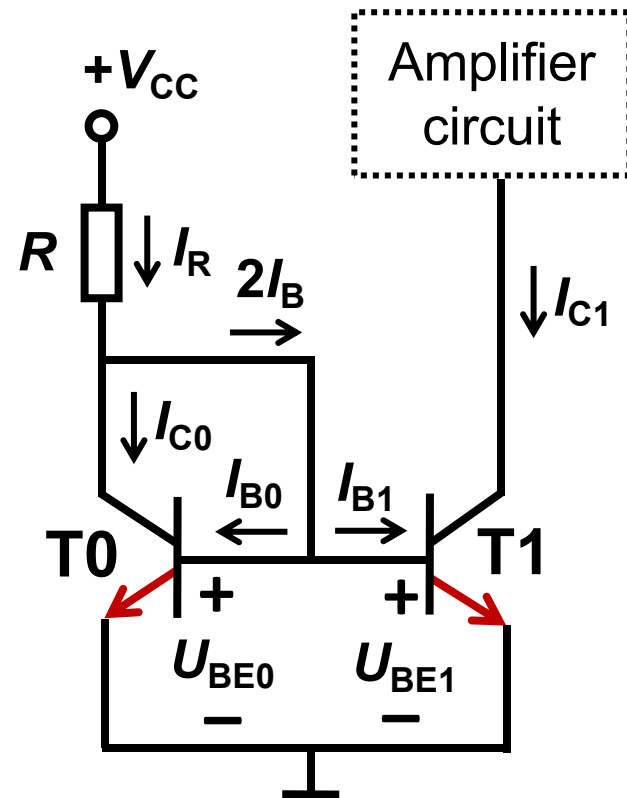
$$I_R = \frac{V_{CC} - U_{BE}}{R}$$

$$\therefore I_R = I_C + 2I_B$$

$$\therefore I_C = \frac{\beta}{\beta + 2} I_R$$

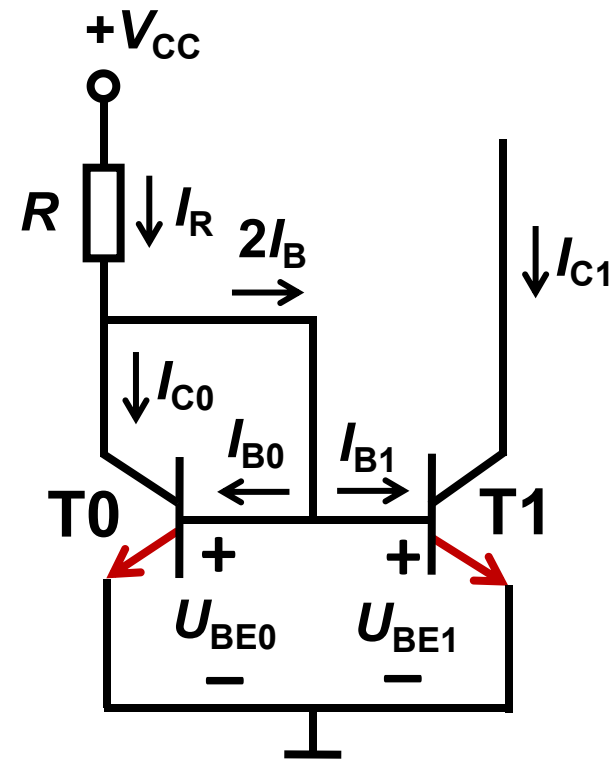
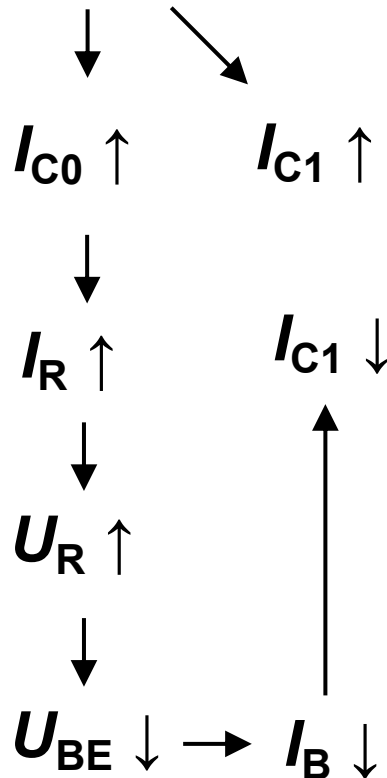
Because $\beta \gg 2$,

$$\therefore I_C \approx I_R = \frac{V_{CC} - U_{BE}}{R}$$



The current mirror has temperature compensation effect 温度补偿作用

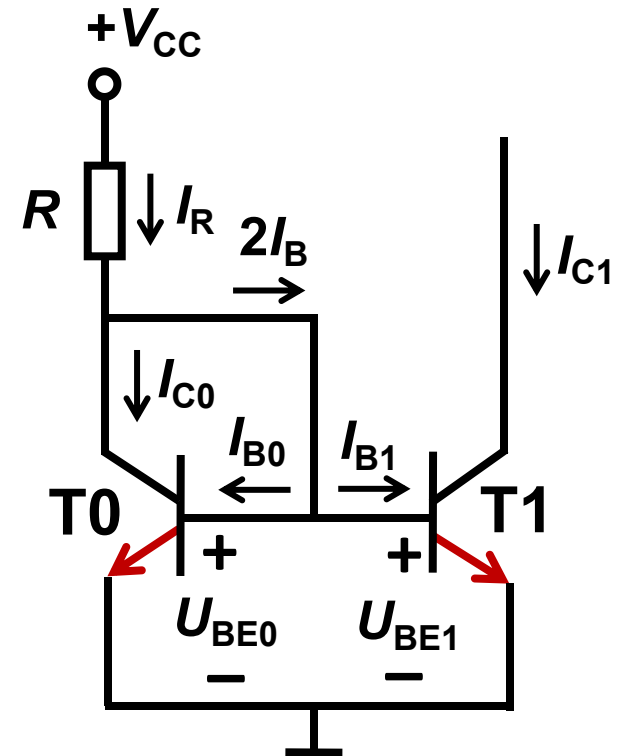
Temperature $\uparrow$



Merit: the current mirror is very simple and widely used

Drawback: For very high current I_C , the power consumption on resistor R is very high

$$I_C \approx I_R = \frac{V_{CC} - U_{BE}}{R}$$



4.4.3 Scaling current mirror 比例电流源

T0 and T1 are identical

$$\beta_0 = \beta_1 = \beta$$

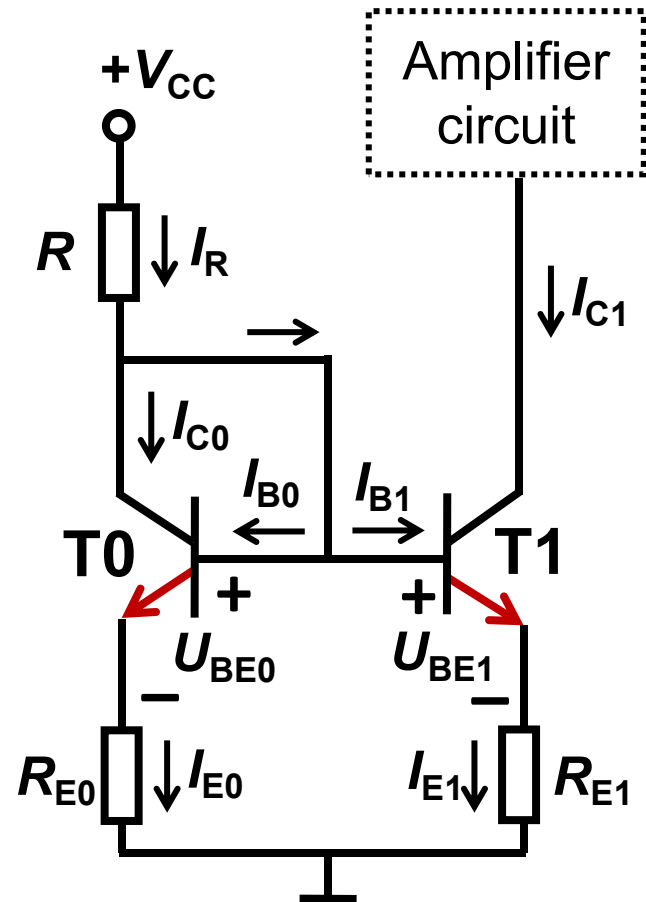
$$\therefore U_{CE0} = U_{BE0}$$

$\therefore$ T0 is always working in amplification region

$$I_{C0} = \beta I_{B0}$$

$$\therefore I_{C0}, I_{E0} \gg I_{B0}, I_{B1}$$

$$I_R \approx I_{E0} \approx I_{C0} \approx \frac{V_{CC} - U_{BE0}}{R_{E0} + R}$$



$$I_{E0}R_{E0} + U_{BE0} = I_{E1}R_{E1} + U_{BE1}$$

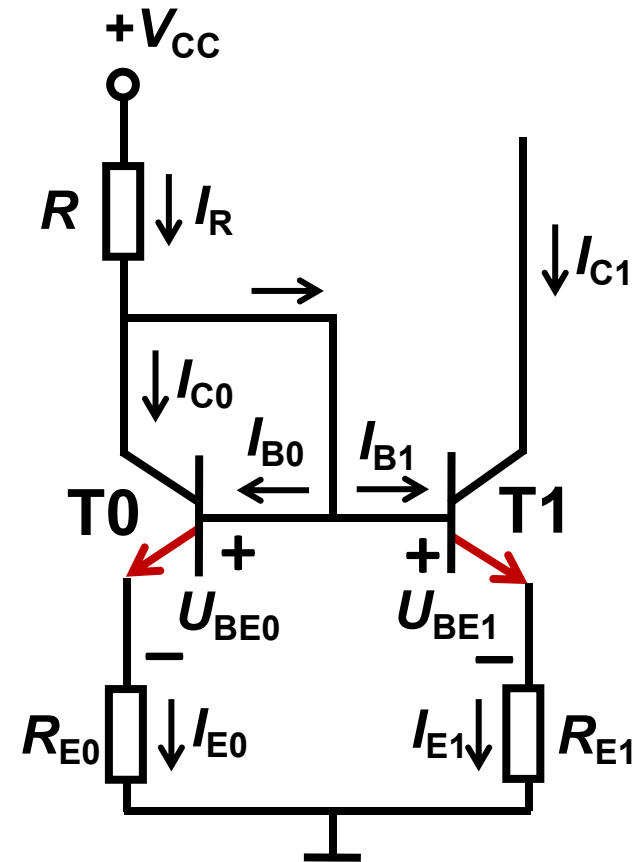
$$\because U_{BE0} \approx U_{BE1}$$

$$\because I_R \approx I_{E0}, I_{C1} \approx I_{E1}$$

$$\therefore I_R R_{E0} = I_{C1} R_{E1}$$

$$I_{C1} = \frac{R_{E0}}{R_{E1}} I_R$$

Through changing the resistance of R_{E0} and R_{E1} , we can increase or decrease I_{C1}



4.4.4 Small source current mirror 微电流源

$$I_{C1} \approx I_{E1} = \frac{U_{BE0} - U_{BE1}}{R_{E1}}$$

Output DC current I_{C1} is very small

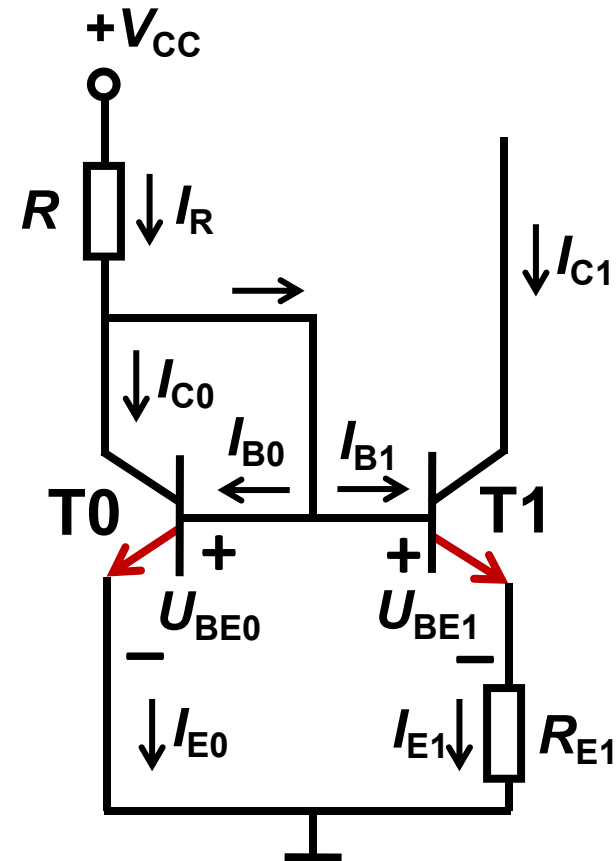
The reference current:

$$I_R = \frac{V_{CC} - U_{BE0}}{R}$$

$$\because I_{C0}, I_{E0} \gg I_{B0}, I_{B1}$$

$$\therefore I_R \approx I_{C0}$$

Invented by Bob Wildar



$$\therefore U_{BE0} = I_{E1}R_E + U_{BE1}$$

$$I_{C1} = I_S \exp\left(\frac{U_{BE1}}{U_T}\right)$$

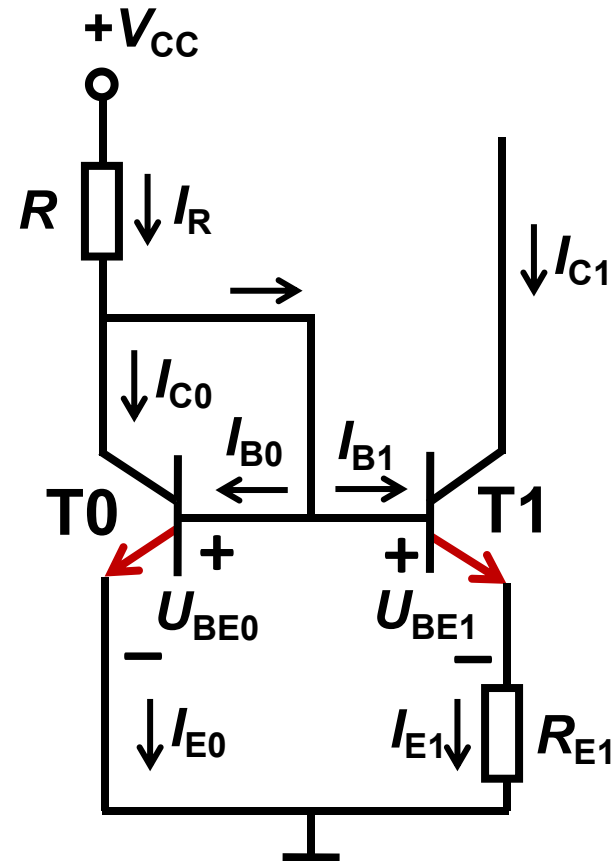
$$U_{BE1} = U_T \ln\left(\frac{I_{C1}}{I_S}\right)$$

$$U_{BE0} = U_T \ln\left(\frac{I_{C0}}{I_S}\right)$$

$$\therefore U_T \ln\left(\frac{I_R}{I_S}\right) \approx I_{C1}R_E + U_T \ln\left(\frac{I_{C1}}{I_S}\right)$$

$$R_E \approx \frac{U_T}{I_{C1}} \ln\left(\frac{I_R}{I_{C1}}\right)$$

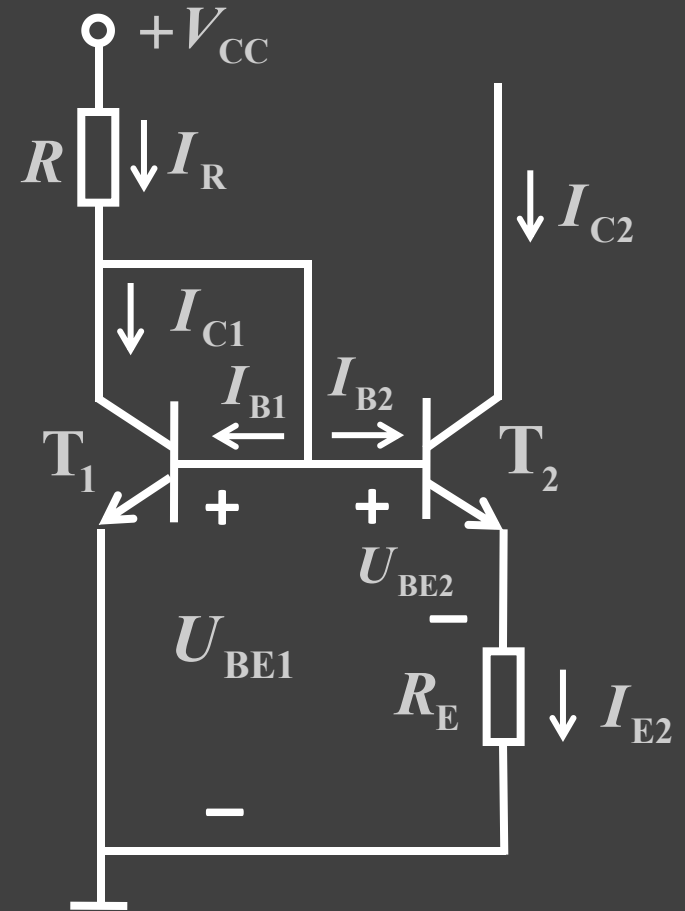
$$U_T = 26 \text{ mV}$$



[1] $I_R = 0.73\text{mA}$, $I_{C2} = 20\mu\text{A}$, ask $R_E = ?$

Ans: $R_E \approx \frac{U_T}{I_{C2}} \ln \frac{I_R}{I_{C2}}$

$$R_E \approx \frac{26}{20 \times 10^{-3}} \ln \frac{0.73}{20 \times 10^{-3}} = 4.86\text{k}\Omega$$



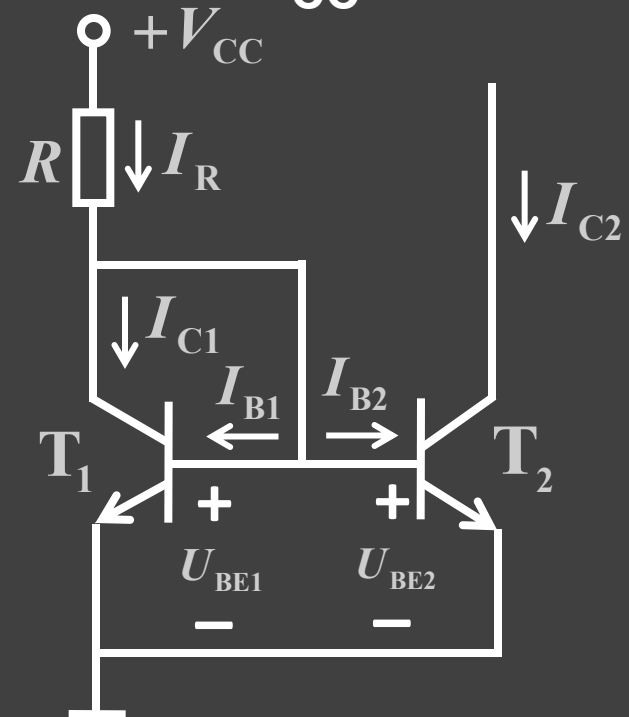
[2] In question [1], if we use current mirror instead of small source current mirror, ask the value of R to achieve $I_{C2}=20\mu\text{A}$, when $V_{CC}=15\text{V}$?

Ans: $I_{C2}=20\mu\text{A}$,

$$I_R=20\mu\text{A}$$

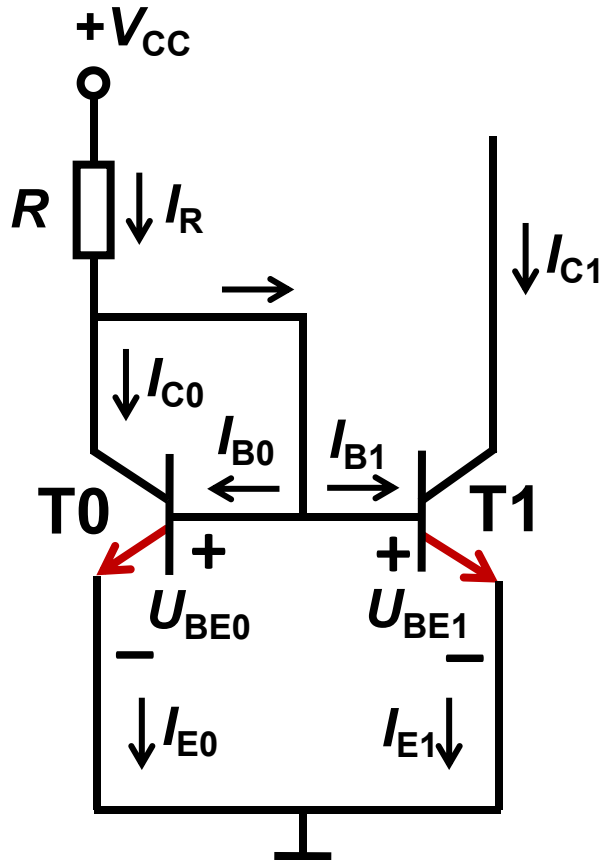
$$R = \frac{V_{CC} - U_{BE1}}{I_R}$$

$$= \frac{15 - 0.7}{0.02} \text{k}\Omega = 715\text{k}\Omega$$



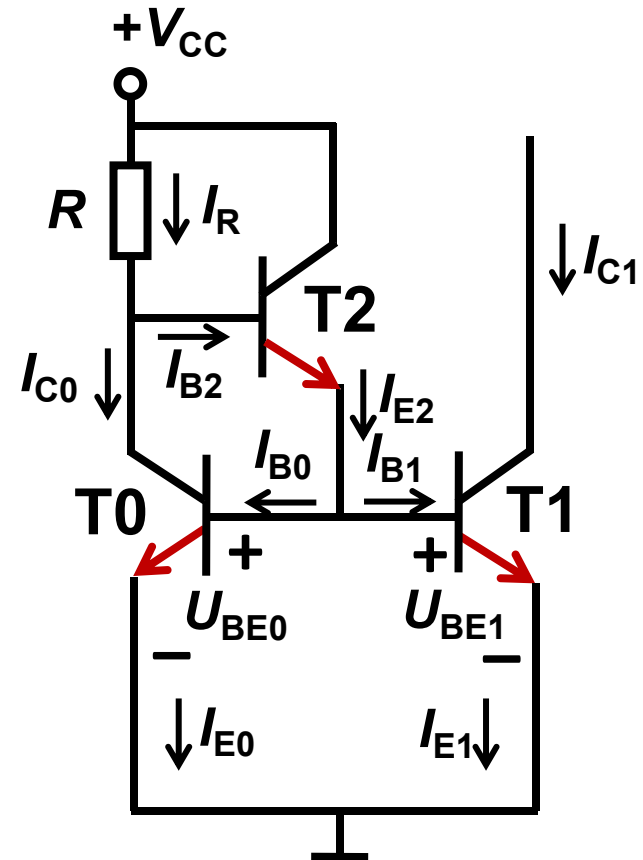
4.4.5 Improved current mirror 改进型电流源

β should be very large



$$I_R = I_{C0} + I_{B0} + I_{B1}$$

β can be small



$$I_R = I_{C0} + (I_{B0} + I_{B1})/(\beta + 1)$$

T0, T1 and T2 are identical

$$\beta_0 = \beta_1 = \beta_2 = \beta$$

$$\therefore U_{BE0} = U_{BE1}$$

$$\therefore I_{B0} = I_{B1} = I_B$$

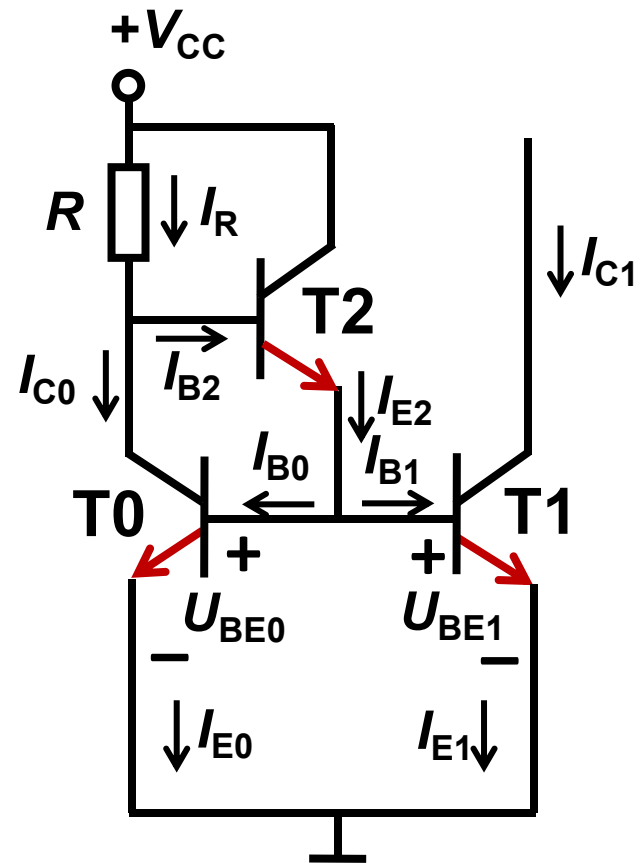
$$\therefore I_{C0} = I_{C1} = I_C$$

$$I_{C1} = I_{C0} = I_R - I_{B2}$$

$$= I_R - \frac{I_{E2}}{\beta + 1} = I_R - \frac{2I_B}{\beta + 1}$$

$$= I_R - \frac{2I_{C1}}{\beta(\beta + 1)}$$

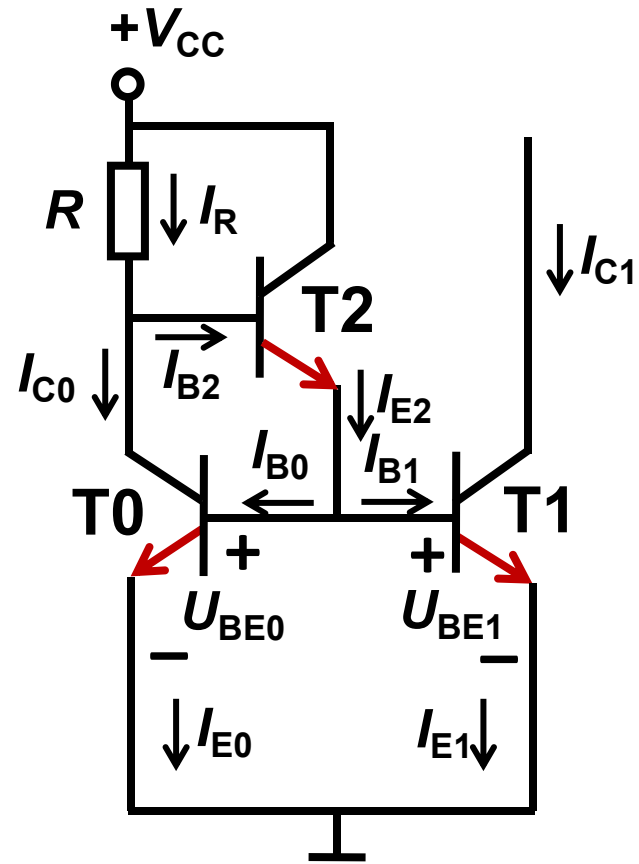
β can be small



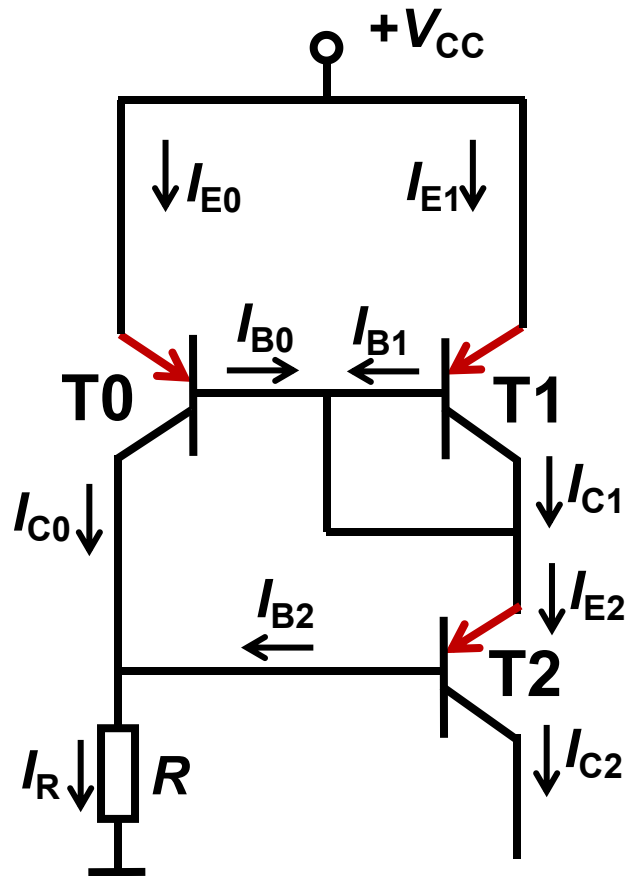
$$I_{C1} = I_R - \frac{2I_{C1}}{\beta(\beta + 1)}$$

$$I_{C1} = \frac{I_R}{1 + \frac{2}{\beta(\beta + 1)}} \approx I_R$$

Even $\beta=10$, $I_{C1}=0.982I_R$



Wilson current mirror



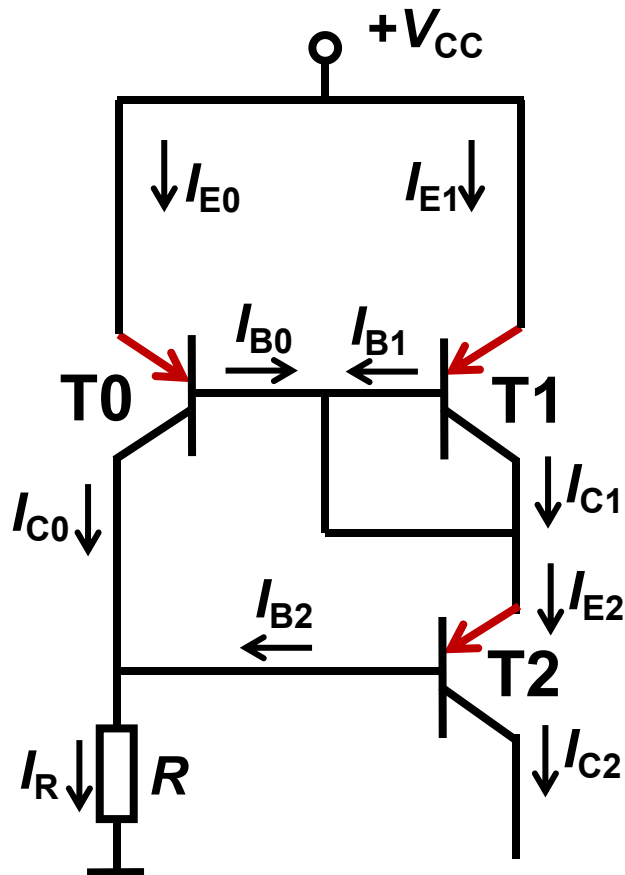
T0, T1 and T2 are identical

$$\beta_0 = \beta_1 = \beta_2 = \beta$$

$$I_{C2} = f(I_R)?$$

1 point

Wilson current mirror



T0, T1 and T2 are identical

$$\beta_0 = \beta_1 = \beta_2 = \beta$$

$$\therefore U_{BE0} = U_{BE1}$$

$$\therefore I_{B0} = I_{B1} = I_B$$

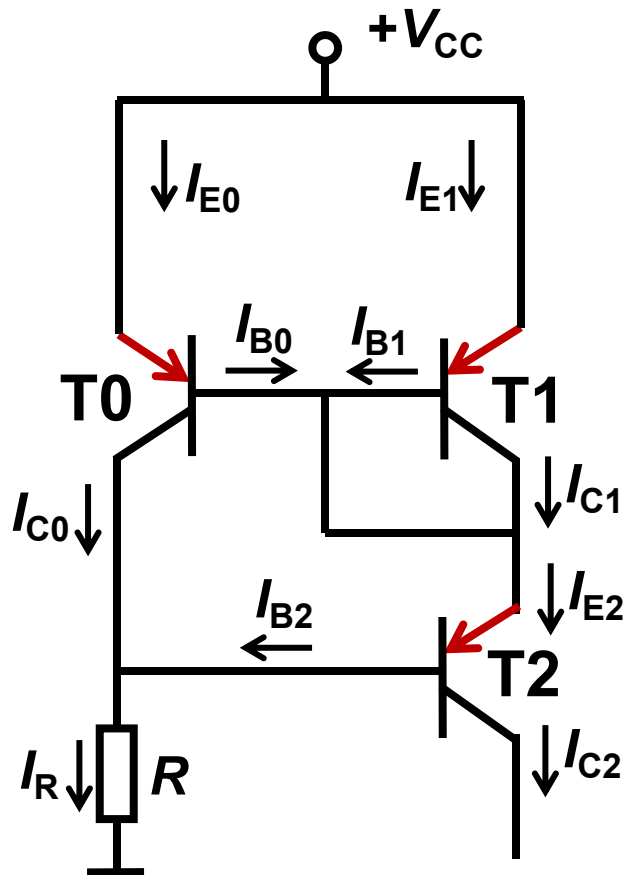
$$\therefore I_{C0} = I_{C1} = I_C$$

$$\therefore I_{E2} = I_C + 2I_B = I_C + \frac{2I_C}{\beta}$$

$$\therefore I_C = \frac{\beta}{\beta + 2} I_{E2} = \frac{\beta + 1}{\beta + 2} I_{C2}$$

$$I_R = I_{B2} + I_C = \frac{I_{C2}}{\beta} + \frac{\beta + 1}{\beta + 2} I_{C2}$$

Wilson current mirror



$$I_{C2} = \left(1 - \frac{2}{\beta^2 + 2\beta + 2}\right) I_R \approx I_R$$

Even $\beta=10$, $I_{C2}=0.984I_R$

Q1: Figure shows the current source of IOC-F007. T10 and T11 are silicon NPN transistor, T12 and T13 are silicon PNP transistor. $\beta=20$, $U_{BE}=0.7$ V. Ask: the collector current of T10, T11, T12 and T13.

Ans: Get the value of I_R first.

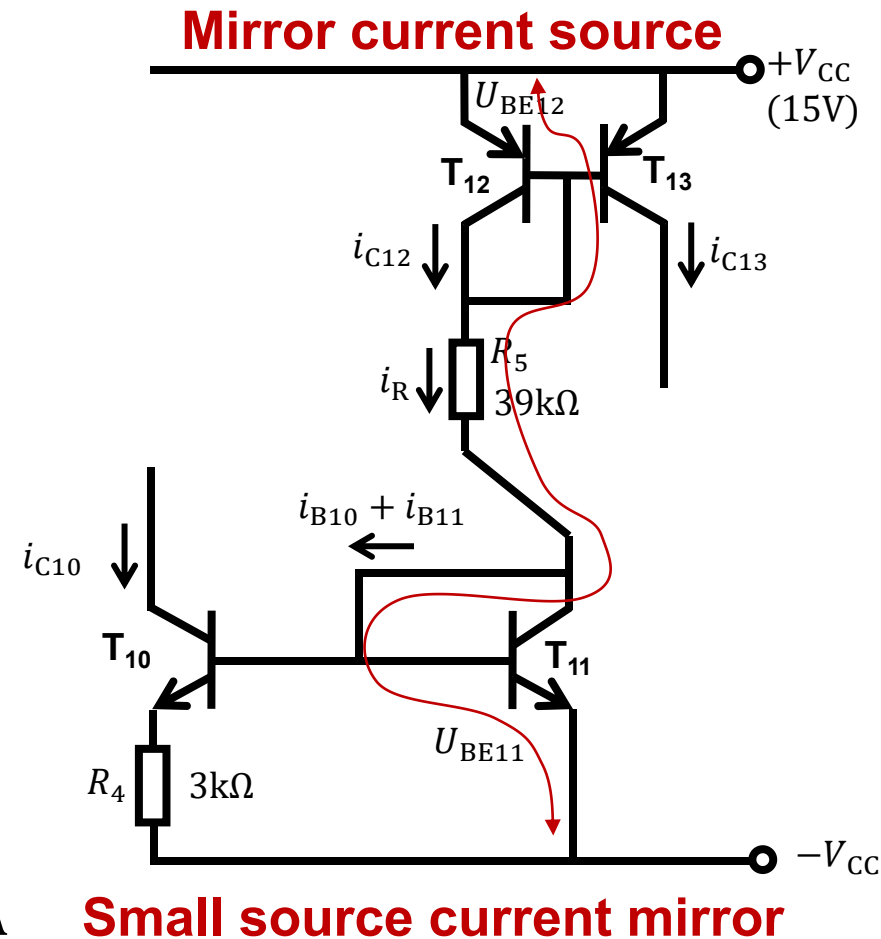
$$I_R = \frac{2V_{CC} - U_{BE11} + U_{BE12}}{R_5}$$

$$= \frac{30 - 0.7 + (-0.7)}{39} \text{ mA}$$

$$\approx 0.73 \text{ mA}$$

$$I_{C10} \approx \frac{U_T}{R_4} \ln \left(\frac{I_R}{I_{C10}} \right) \approx 28 \text{ } \mu\text{A}$$

$$I_{C12} = I_{C13} = \frac{\beta}{\beta + 2} I_R \approx 0.66 \text{ mA}$$

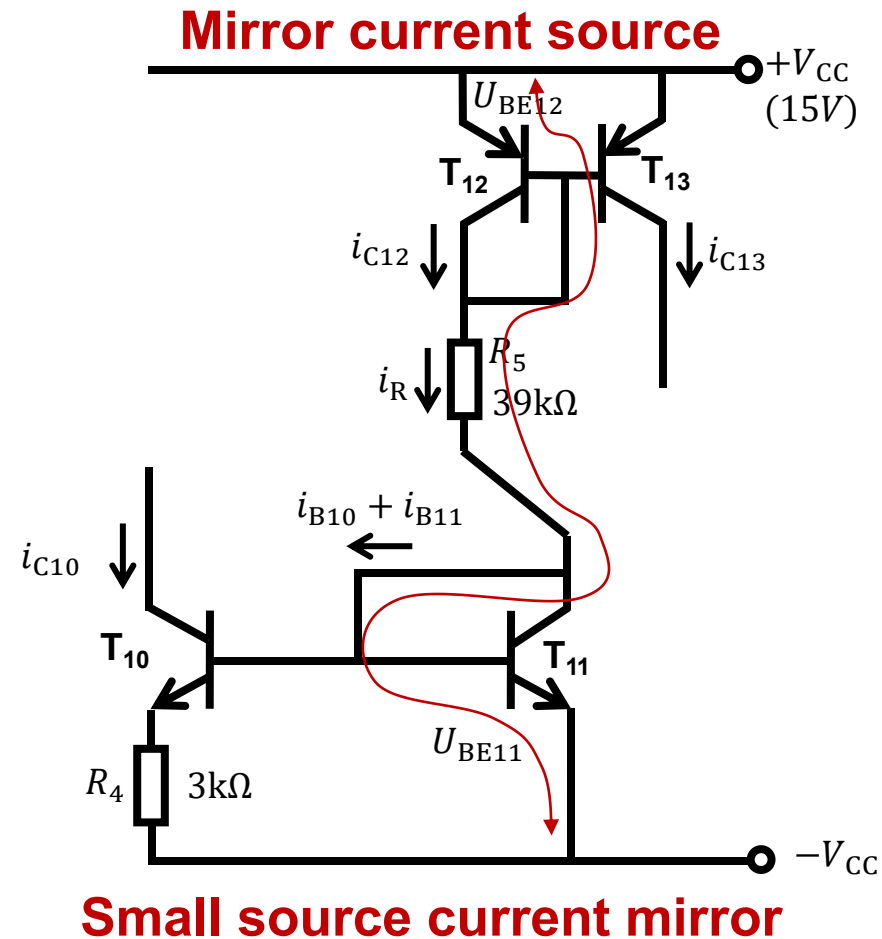


$$I_{C11} = I_R - I_{B10} - I_{B11}$$

$$\approx I_R - I_{B11}$$

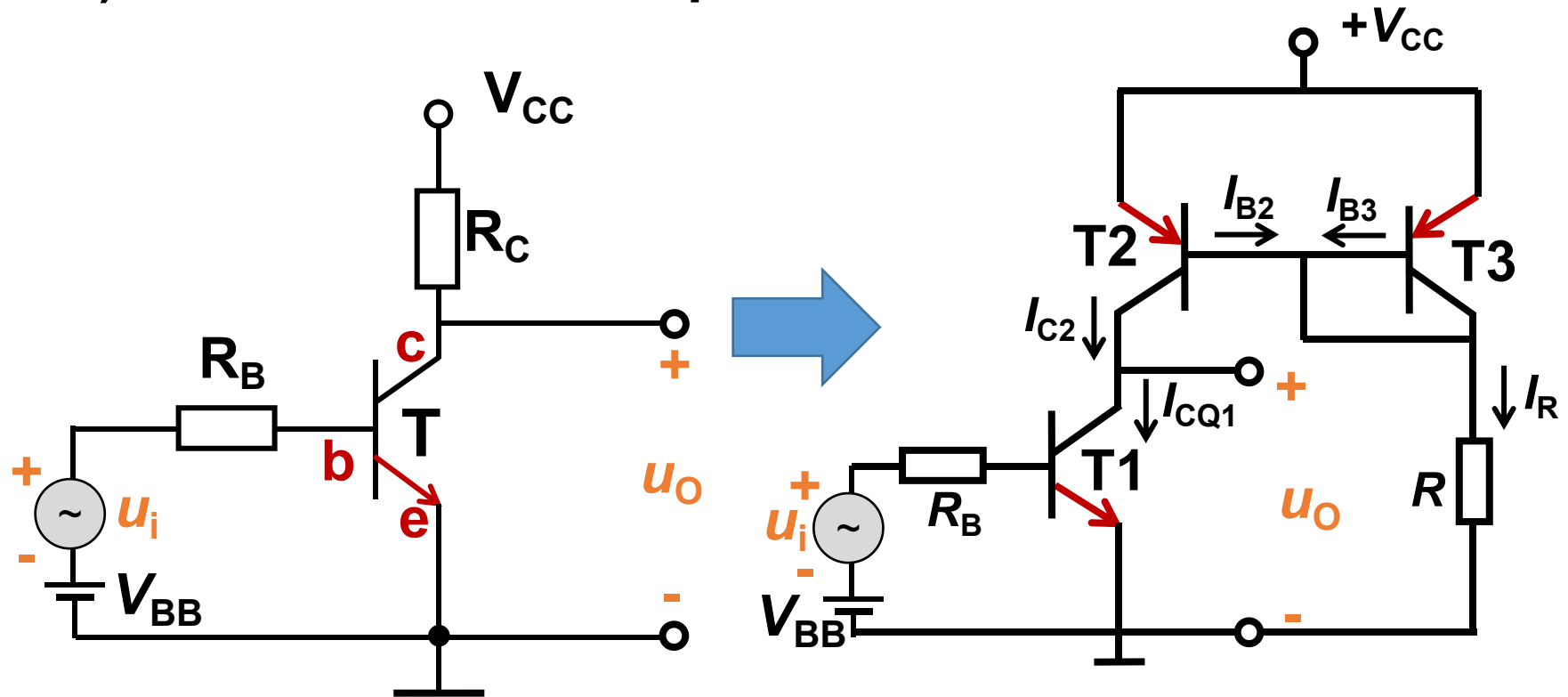
$$= I_R - \frac{I_{C11}}{\beta}$$

$$I_{C11} = \frac{\beta}{\beta + 1} I_R \approx 0.7 \text{ mA}$$



4.4.6 Amplifier circuits with current source

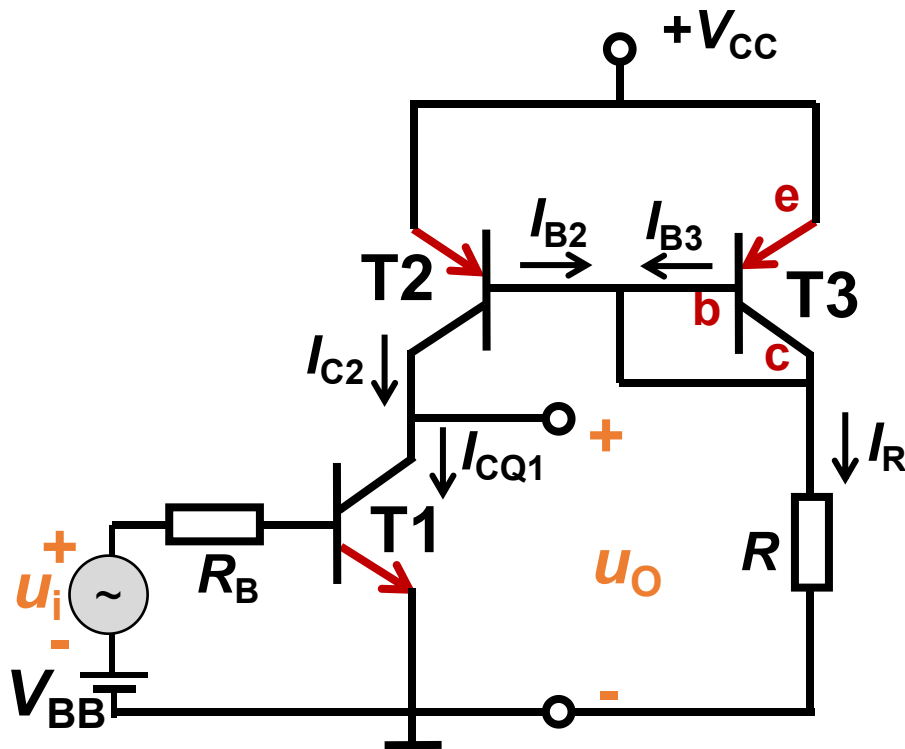
1) Common-emitter amplifier with current mirror



Replace R_C with a current mirror

The current source (acting as a resistor) is also called **active load** 有源负载

What's the value of DC current I_{CQ1} ?



T2 and T3 are identical

$$\beta_2 = \beta_3 = \beta$$

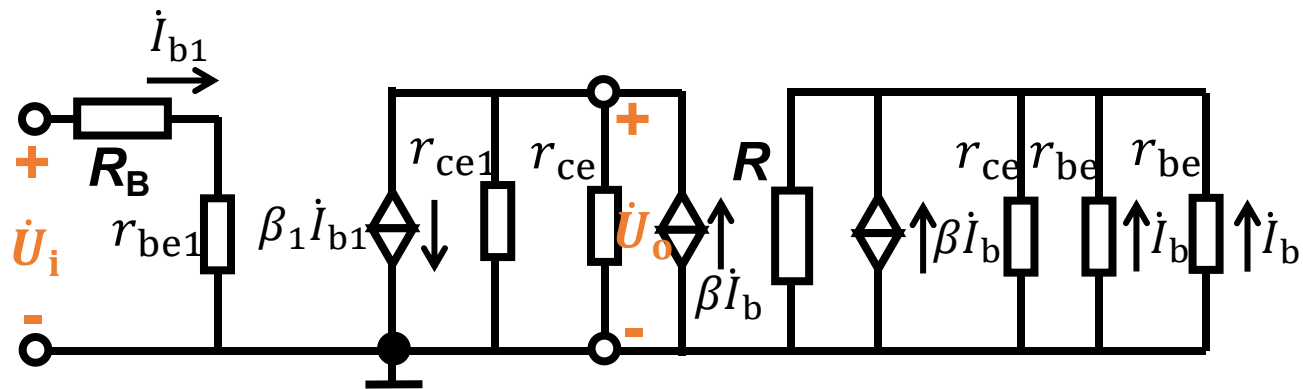
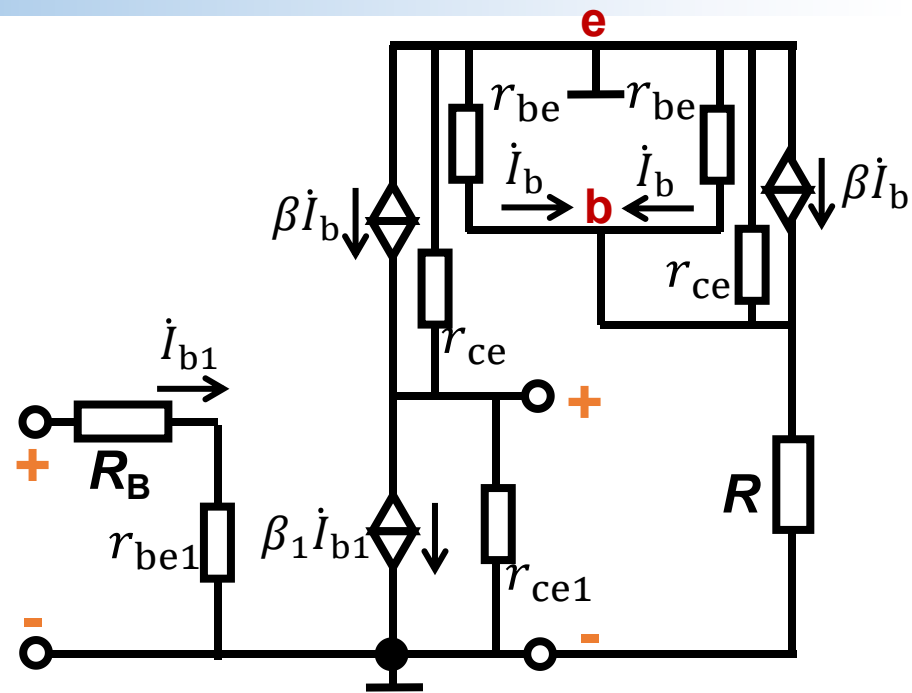
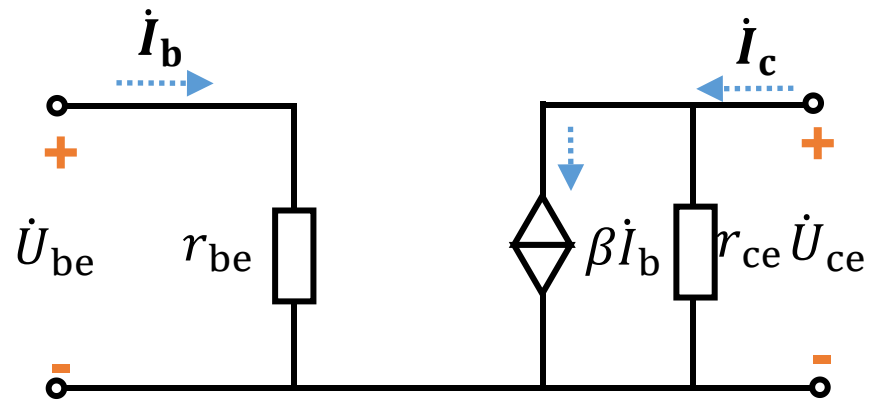
$$I_{B2} = I_{B3} = I_B$$

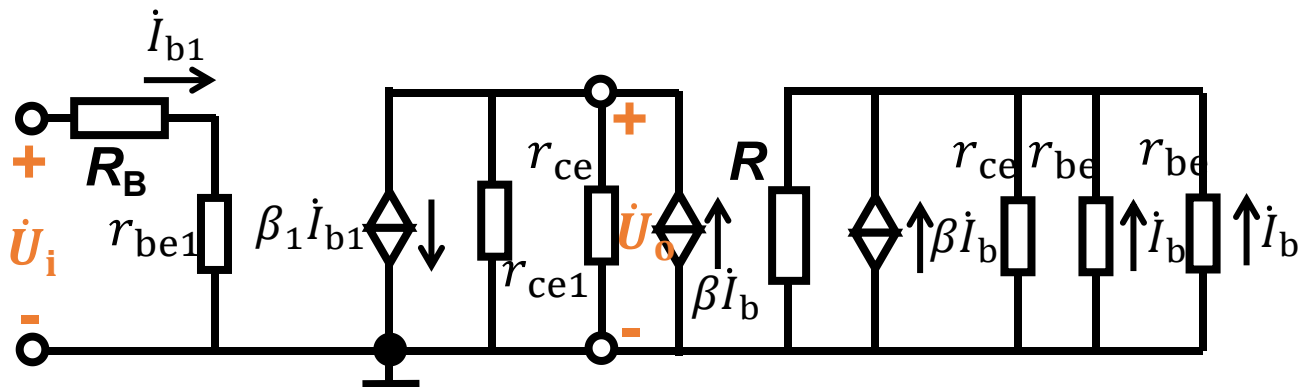
$$I_{C2} = I_{C3} = \beta I_B$$

$$I_R = \frac{V_{CC} - U_{EB3}}{R}$$

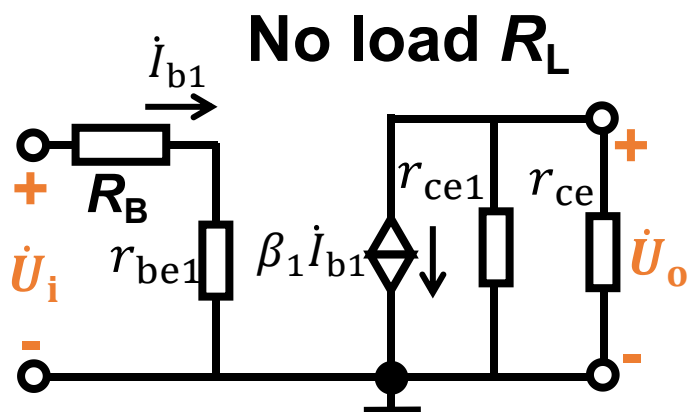
$$I_{CQ1} = I_{C2} = \frac{\beta}{\beta + 2} I_R$$

What's the AC voltage gain?

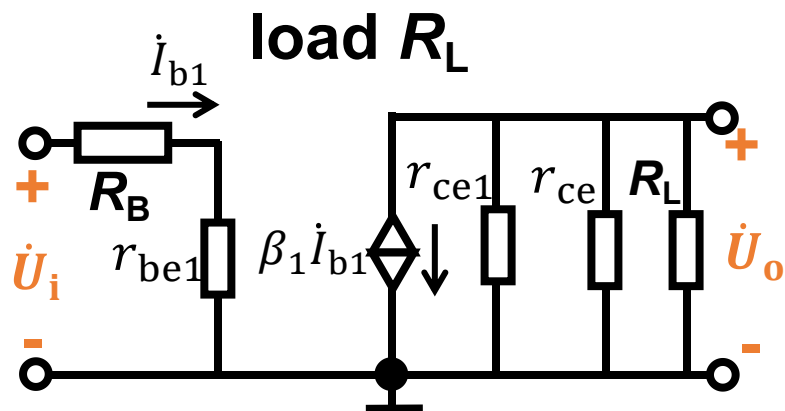




$$\dot{I}_b = 0$$



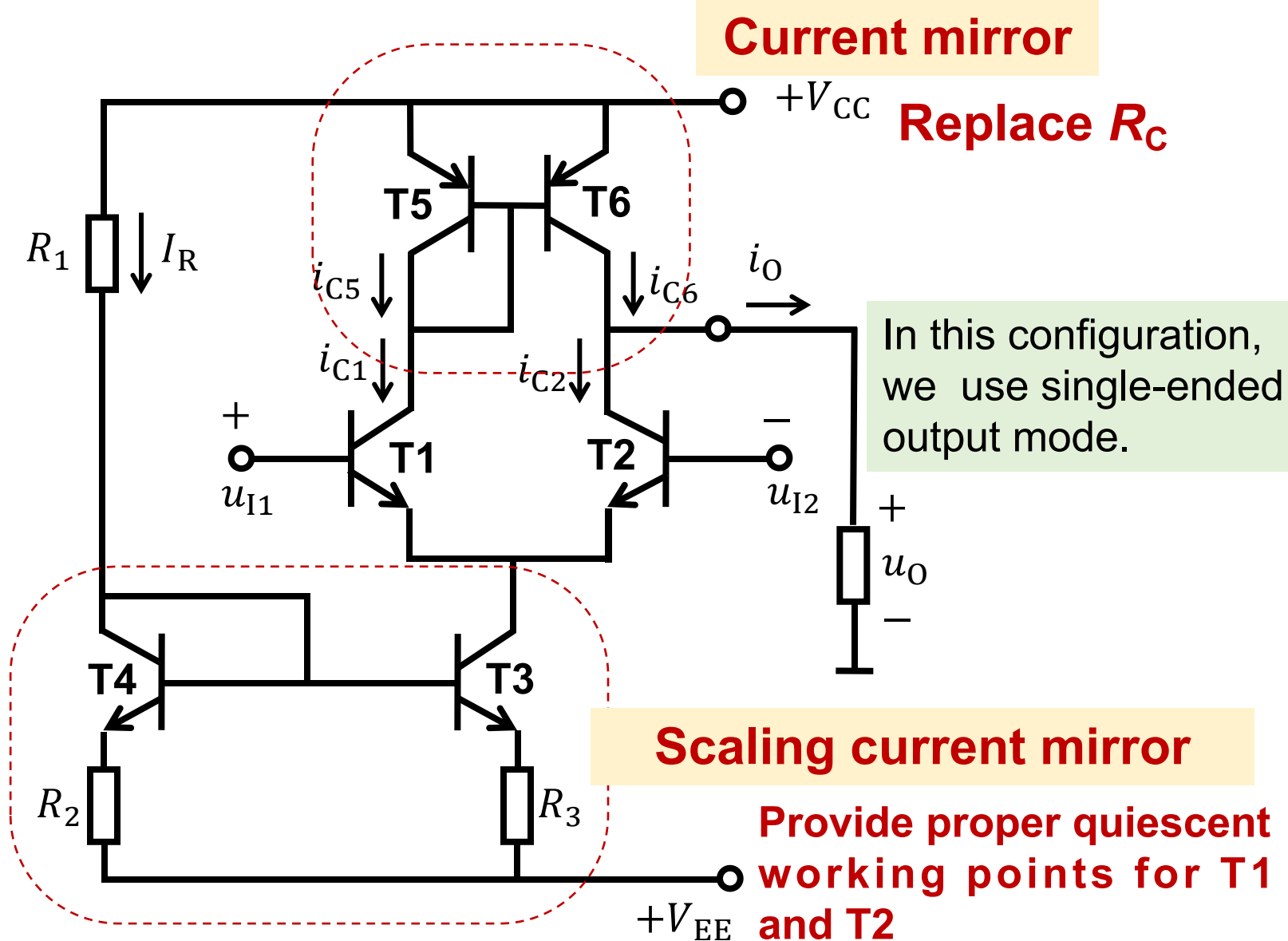
$$\dot{U}_i = \dot{I}_{b1}(R_B + r_{be1})$$



$$\dot{U}_o = -\beta_1 \dot{I}_{b1}(R_L // r_{ce1} // r_{ce})$$

$$\dot{A}_u = -\frac{\beta (R_L // r_{ce1} // r_{ce})}{R_B + r_{be1}}$$

2) Differential amplifier with current source



Analyze AC voltage gain

T5 and T6 are identical

$$\therefore i_{c5} = i_{c6}$$

$$u_{i1} = u_{i2} \text{ (common - mode)}$$

$$i_{c2} = i_{c1}$$

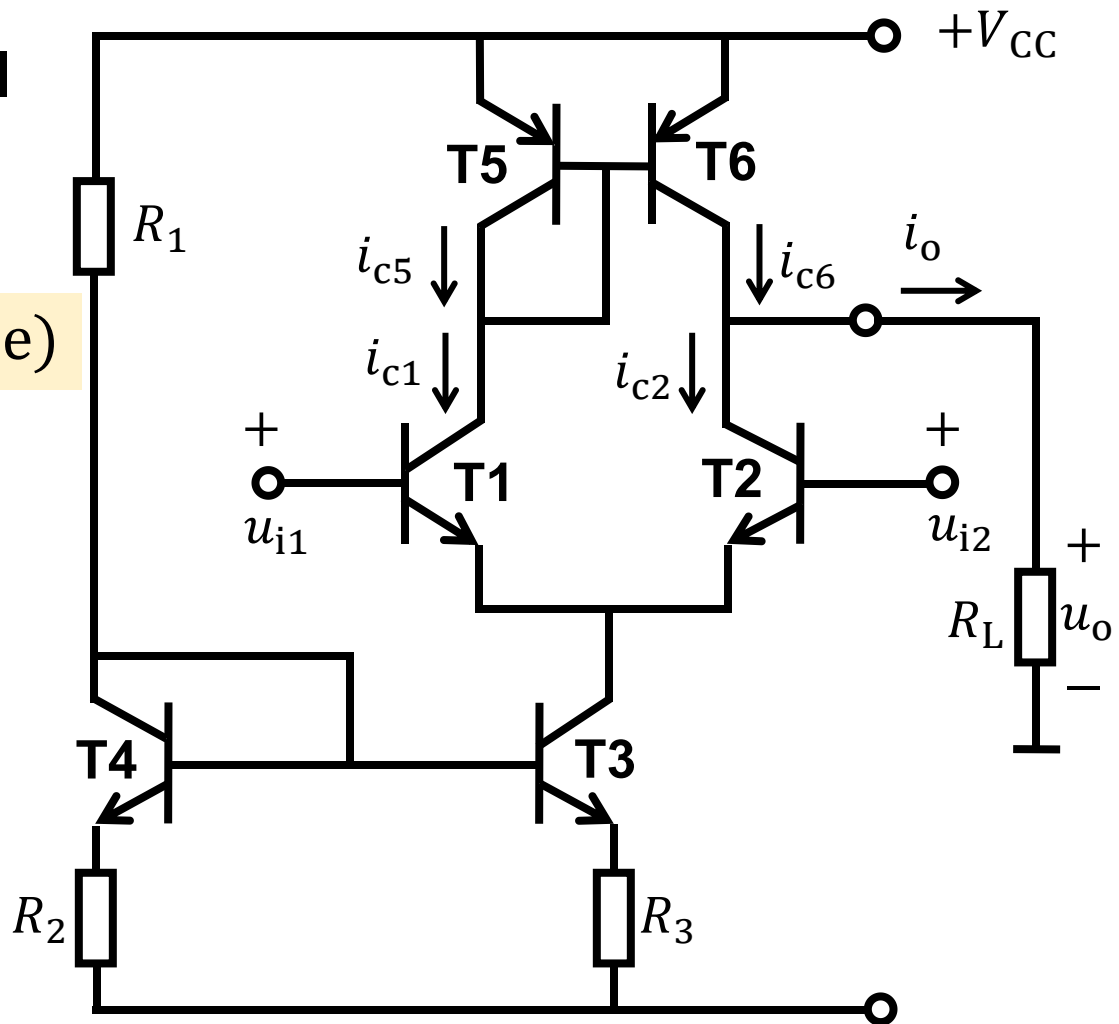
$$\therefore i_{c5} \approx i_{c1}$$

$$\therefore i_o = i_{c6} - i_{c2}$$

$$= i_{c5} - i_{c1} \approx 0$$

Voltage gain for common mode is almost zero.

Instantaneous value 瞬时值



$$u_{i1} = -u_{i2}$$

(differential – mode)

$$i_{c2} = -i_{c1}$$

$$\therefore i_o = i_{c6} - i_{c2} \approx 2i_{c1}$$

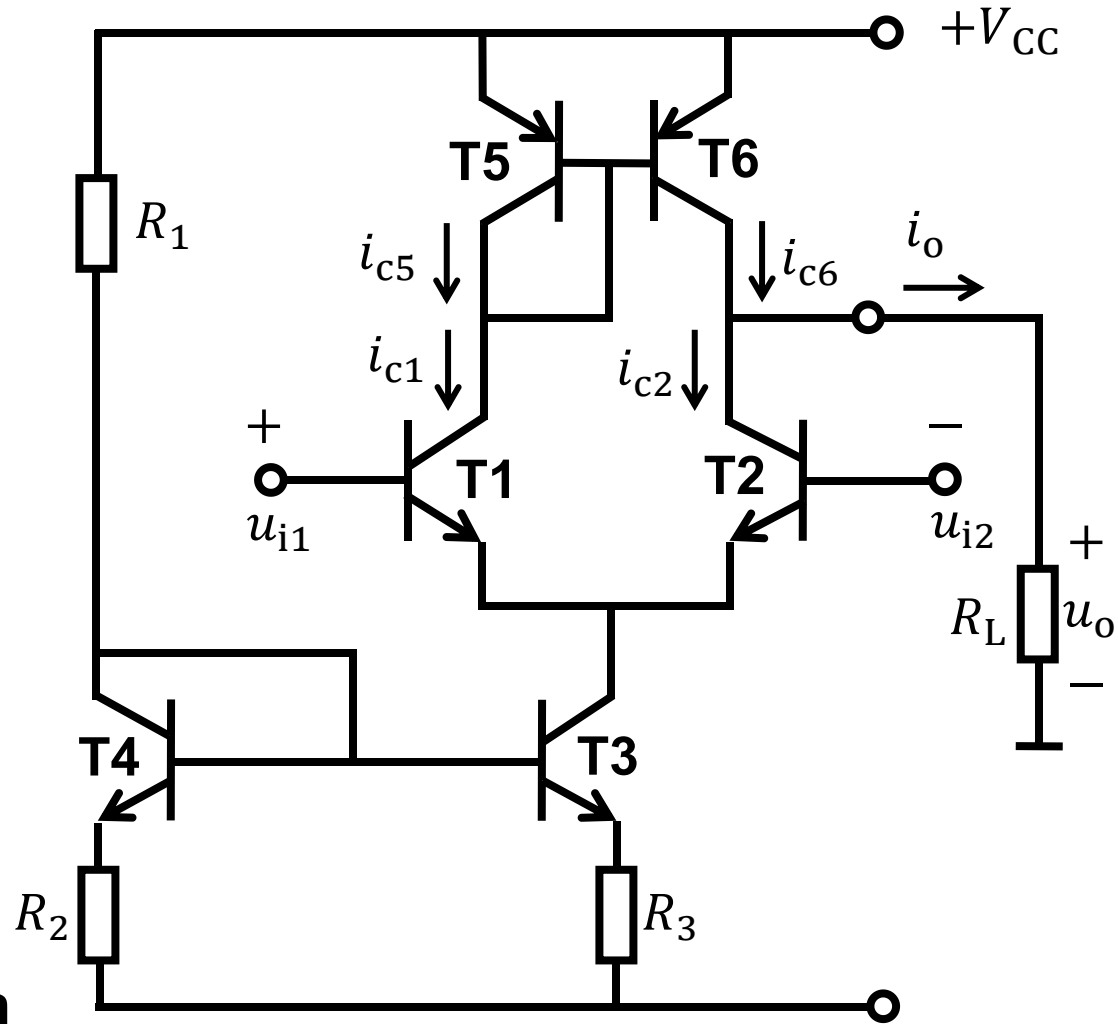
$$u_i = 2u_{i1} = 2i_{b1}r_{be1}$$

$$u_o = i_o R_L \approx 2i_{c1} R_L$$

Voltage gain for differential mode:

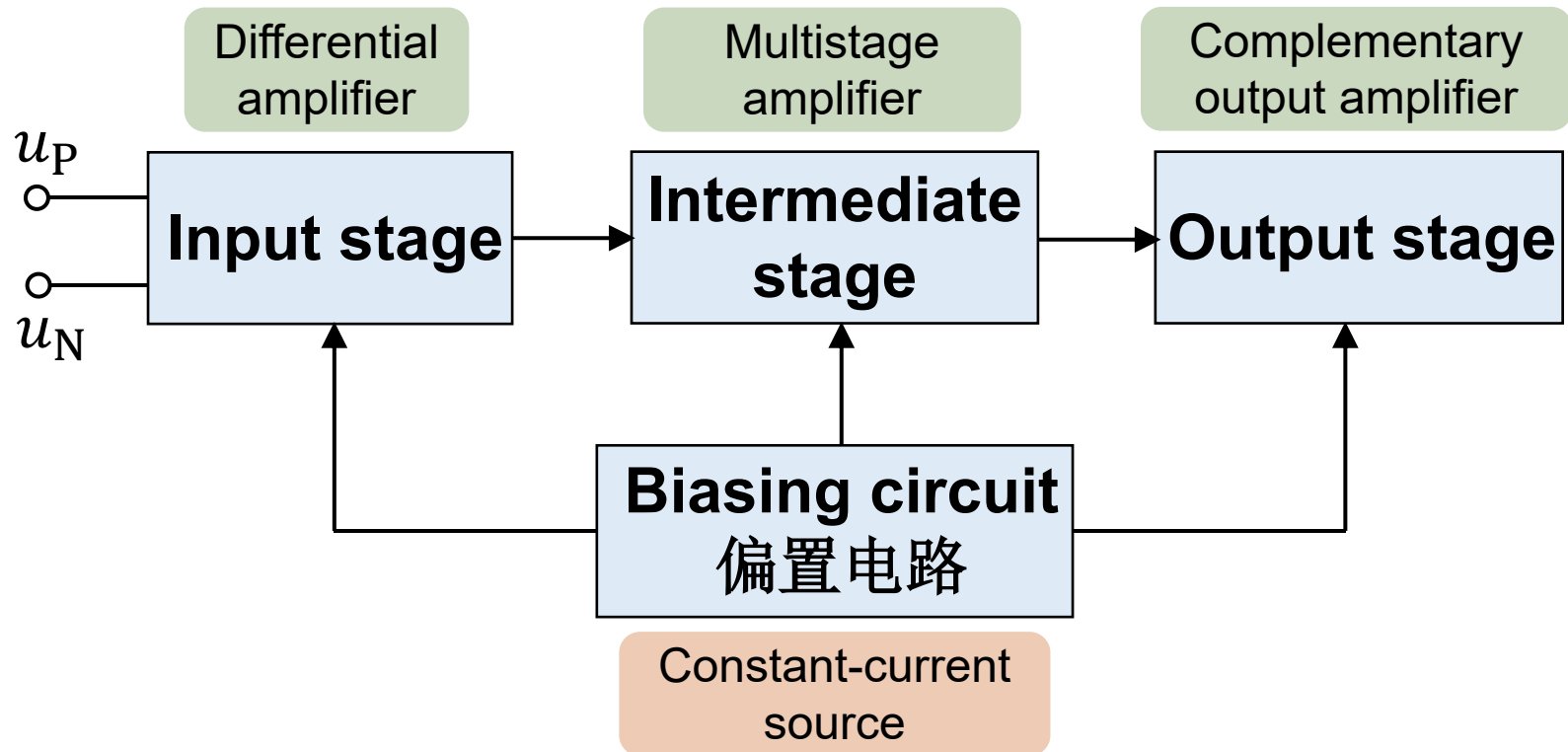
$$A_{ud} = \frac{u_o}{u_i} \approx \frac{\beta R_L}{r_{be1}}$$

Double the voltage gain of single-ended output using current mirror



Summary

The structure of the integrated operational circuit (IOC)



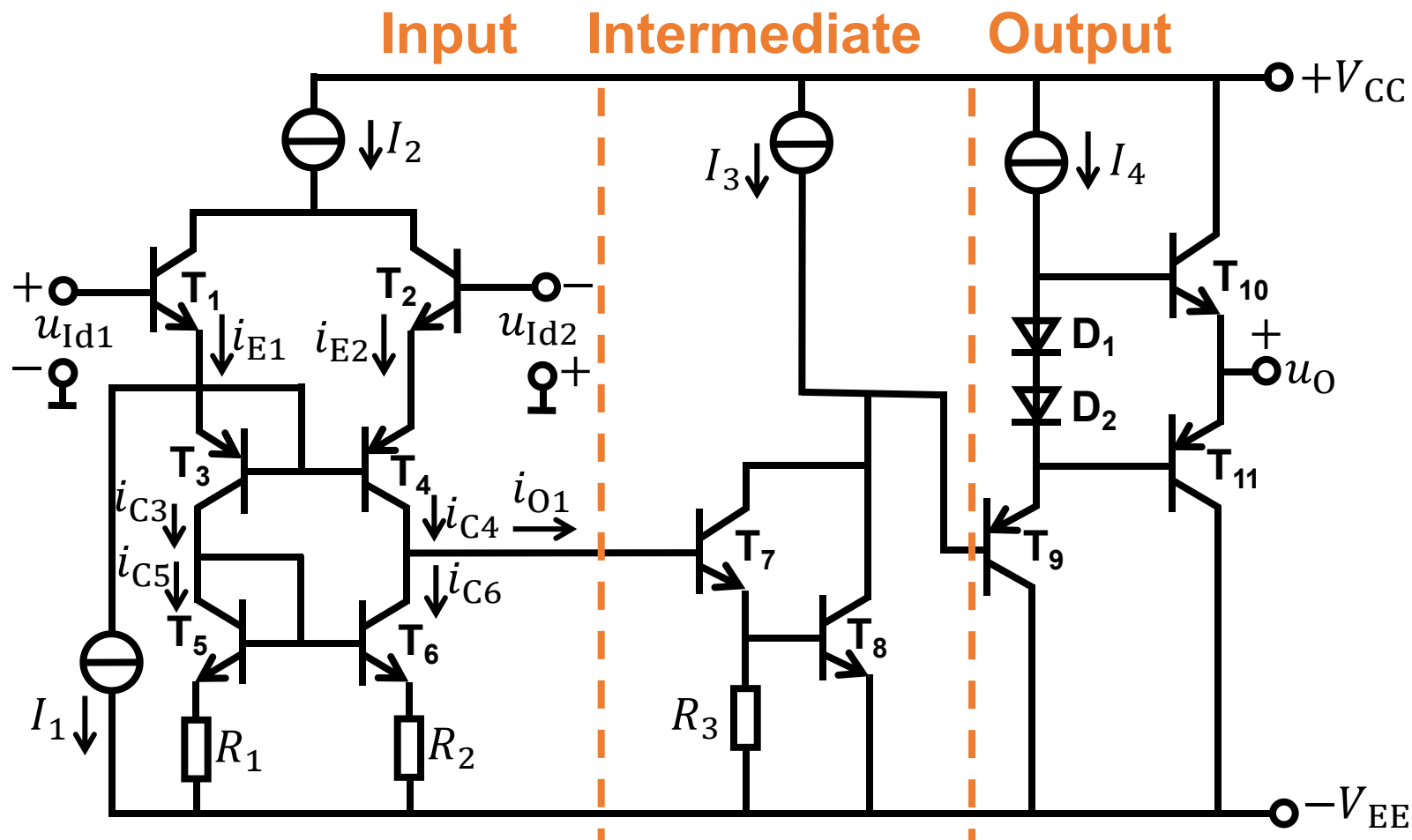
Homework 4-2: refer to QQ group and blackboard

Hand in together with homework 4-1

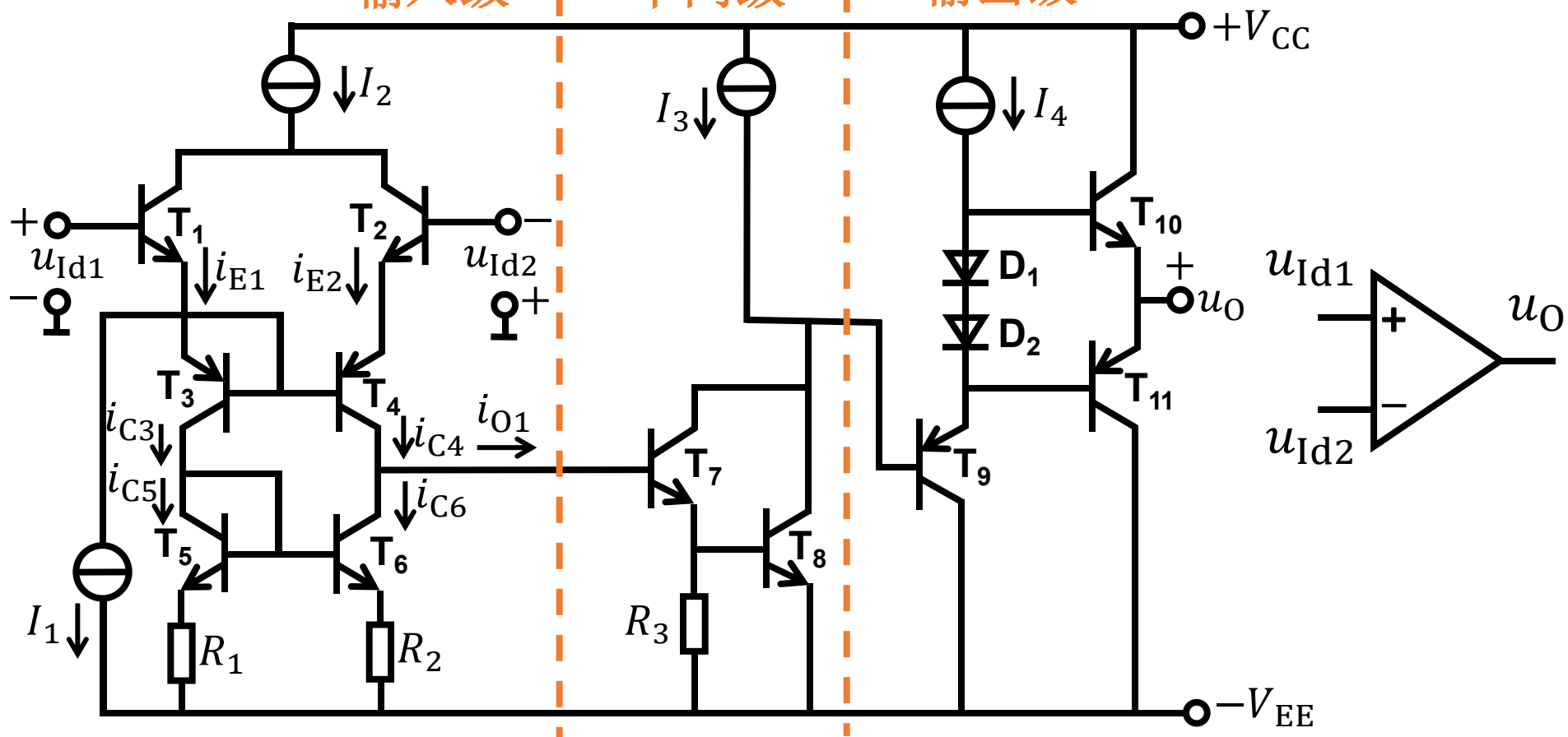
Deadline: 2025-04-23 11:59pm

4.5 Integrated operational amplifier 集成运算放大器

Bipolar general integrated operational amplifier 双极型通用集成运算放大器



Input Intermediate Output
输入级 中间级 输出级



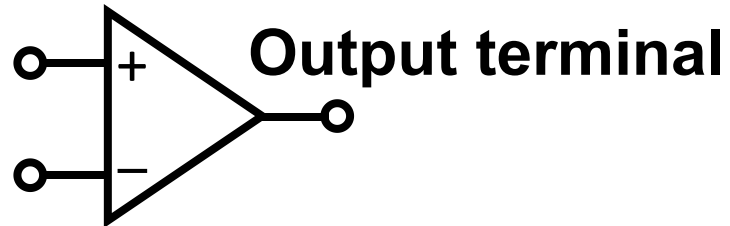
Electronic symbol of integrated operational amplifier 集成运放符号

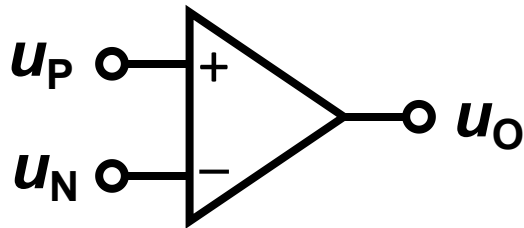
Non-inverting input terminal

同相输入端

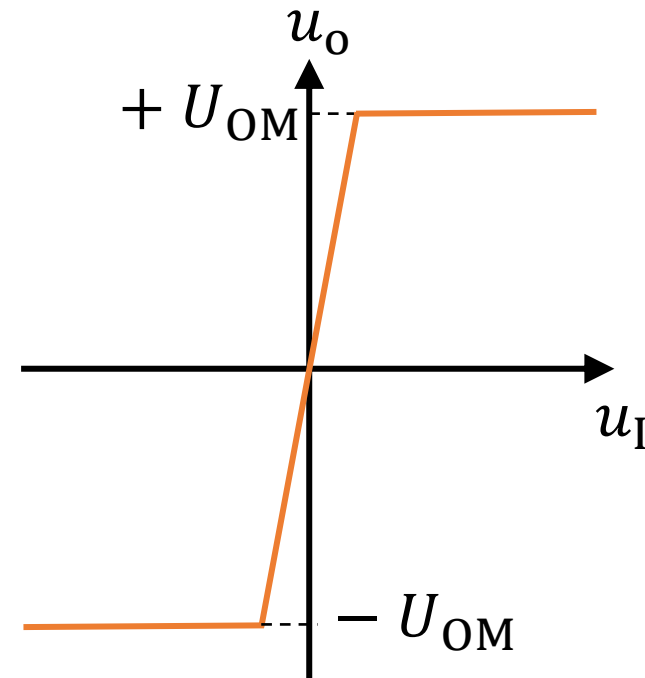
反相输入端

Inverting input terminal





Input signal $u_{Id} = u_P - u_N$

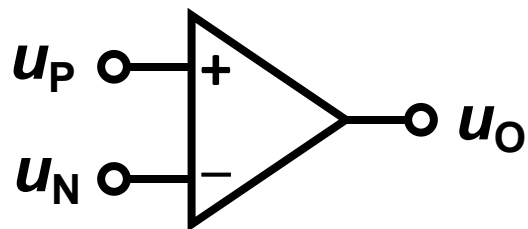


Linear amplification region 线性放大区

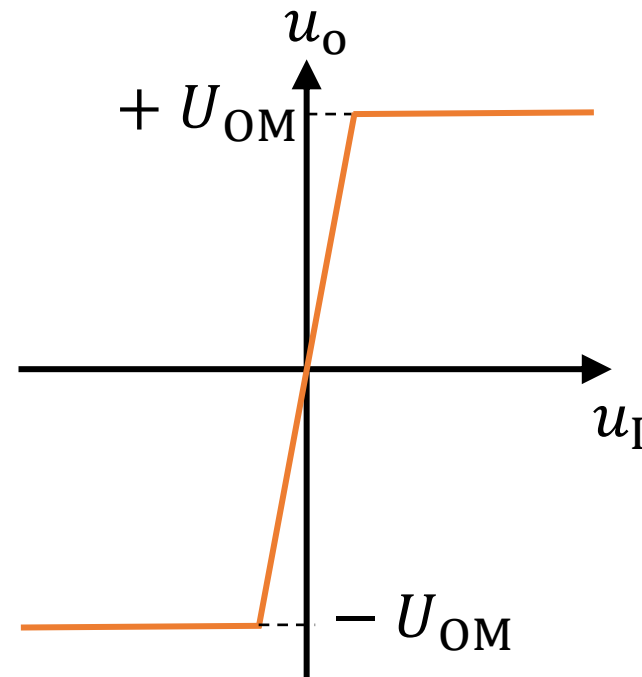
$$u_O = A_{od}(u_P - u_N)$$

A_{od} : Open loop differential mode voltage gain 开环差模电压增益

In IOC, A_{od} can be $>100,000$



$$u_{Id} = u_P - u_N$$



Saturation region 饱和区

$$u_O = +U_{OM} \text{ or } -U_{OM}$$

If $U_{OM}=14 \text{ V}$ and $A_{od}=500,000$, integrated operational amplifier enters saturation region when $|u_P - u_N| > 28 \mu\text{V}$

The equivalent model at low frequency

